

Integrity Management, Monitoring and Maintenance of Eni UK CCS Pipelines

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1 ABSTRACT

Carbon Capture and Storage (CCS) is a technology for mitigating climate change by capturing CO₂ emissions from industrial processes and transporting this CO₂ through pipelines to underground geological formations, often depleted oil and gas reservoirs or saline aquifers, to provide secure storage for the CO₂.

Eni UK's Liverpool Bay Asset has been selected as a suitable system for CCS. This project covers the design and installation of new pipeline infrastructure as well as the repurposing of existing LBA pipelines. CO₂ will be transported from the emitters and delivered to the infield platforms via the onshore and offshore pipelines and will be injected into the existing reservoirs.

Repurposing of the existing pipelines requires careful consideration of safety, capacity, and compatibility with current operations. Collaboration between industries, government, and regulators is essential to ensure proper infrastructure usage, emission reduction goals, and regulatory compliance.

Introducing CO₂ into a pipeline network poses several threats to carbon steel pipelines, particularly when any of the following contaminants are also present:

- Water Content
- Existing Debris
- Corrosion, subsequent corrosion products and pitting corrosion.

A recent (2020/2023) Inline Inspection (ILI) programme has identified residual solid debris in the six gas pipelines that will be repurposed for CO₂ service. Debris will readily permit water hold-up to occur within the pipeline. It is critical that all debris and water is first removed. A suitable pipeline cleaning and drying technique must be utilised to achieve a very low dew point.

e-Splora, video inspection tools and other available techniques will be utilised to evaluate the quality of the pipeline cleanliness prior to commencing CO₂ service. The e-Splora (proprietary Eni technology) is a suite of advanced technologies developed by Eni for pipeline inspection and maintenance. It includes various smart devices designed to detect and localize anomalies in pipelines, ensuring their integrity and reducing maintenance costs. The e-Splora multiparametric sensor can be deployed directly into pipelines by attaching it to pigs to collect data on pressure, temperature, and other critical parameters.

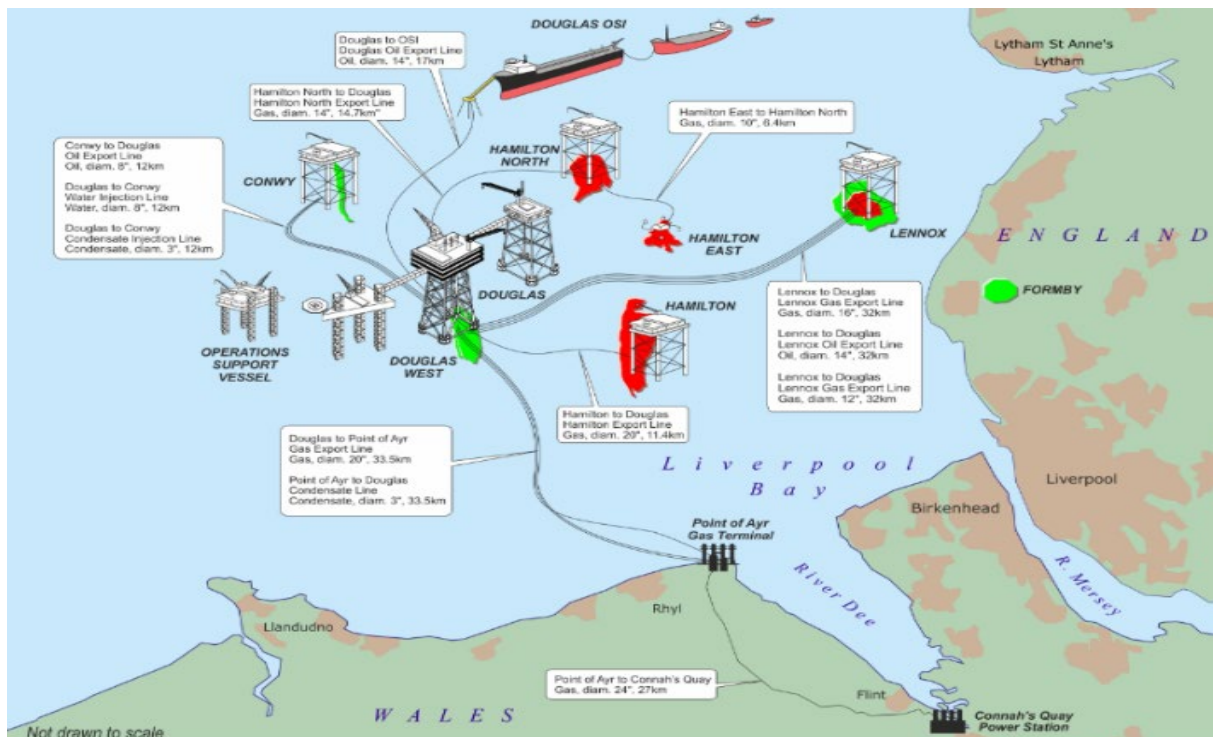
Pipeline operators use various integrity management strategies, which include monitoring and maintenance programs to detect and address corrosion and other integrity concerns during the pipeline design life. Magnetic flux leakage (MFL) inspection has been utilised to identify pipeline threats such as corrosion, defects, damage, etc. to assess the pipeline condition. This paper considers MFL technology use, supporting the repurposing and future integrity, monitoring, and maintenance of the new and existing pipelines.

2 INTRODUCTION

The Liverpool Bay Facilities operated by Eni UK Liverpool Bay Asset (Eni UK LBA) consist of a central manned processing platform (Douglas), four unmanned satellite platforms (Lennox, Hamilton, Hamilton North, Conwy), a floating Oil Storage Installation (OSI), the Point of Ayr Gas Terminal and the interconnecting pipelines and main sales gas pipeline to Connah's Quay power station.

The Douglas complex (53° 32' 17" N, 0.3° 34' 32" W) is located approximately 27km off the North Wales coast and 32km off the Lancashire coast in 27m (Lowest Astronomical Tide, LAT) of water. The Eni owned Liverpool Bay installations were commissioned in 1995/6. The design life of these structures and plant is nominally 30 years.

Figure 1 - Eni UK Liverpool Bay Existing Pipelines Facilities



In the pursuit of sustainable solutions to address climate change, Carbon Capture and Storage (CCS) is an essential component for the UK's energy transition towards net zero. The United Kingdom has an existing capacity to safely store up to 78 billion tonnes of CO₂ under its seabed [1]. Therefore, repurposing of existing oil and gas pipelines for the transportation phase of CCS emerges as a promising strategy.

Eni UK Liverpool Bay Asset has been selected as a suitable system for CCS. This project covers the design and installation of new pipeline infrastructure as well as the repurposing of the existing LBA wet gas pipelines. CO₂ (in gas and later dense phase) will be transported from the emitters and delivered to the infield platforms via the onshore and offshore pipelines and will be injected into the existing reservoirs.

Figure 2 - Eni UK Liverpool Bay Future CCS Storage and Transportation System



Mechanical and chemical cleaning, and subsequent drying will be required to achieve the very low dew point necessary for CO₂ Service. The CCS project will use MFL technology for pipeline condition assessment, e-Splora, video inspection and other available industry techniques to evaluate the quality of the pipeline cleanliness.

Both the existing pipelines and the new pipelines require the use of advanced technologies to ensure their integrity and reliability throughout the process of the project execution, integrity management, and operation as shown in Figure 3.

- Design, Construction, and Installation of new pipelines,
- Integrity Risk Assessment,
- Inspection, Maintenance, and Testing,
- Monitoring and Audit, and
- Emergency Response and Repairs.

Figure 3 – Eni UK CCS Pipelines throughout the process of the project execution, integrity management, and operation



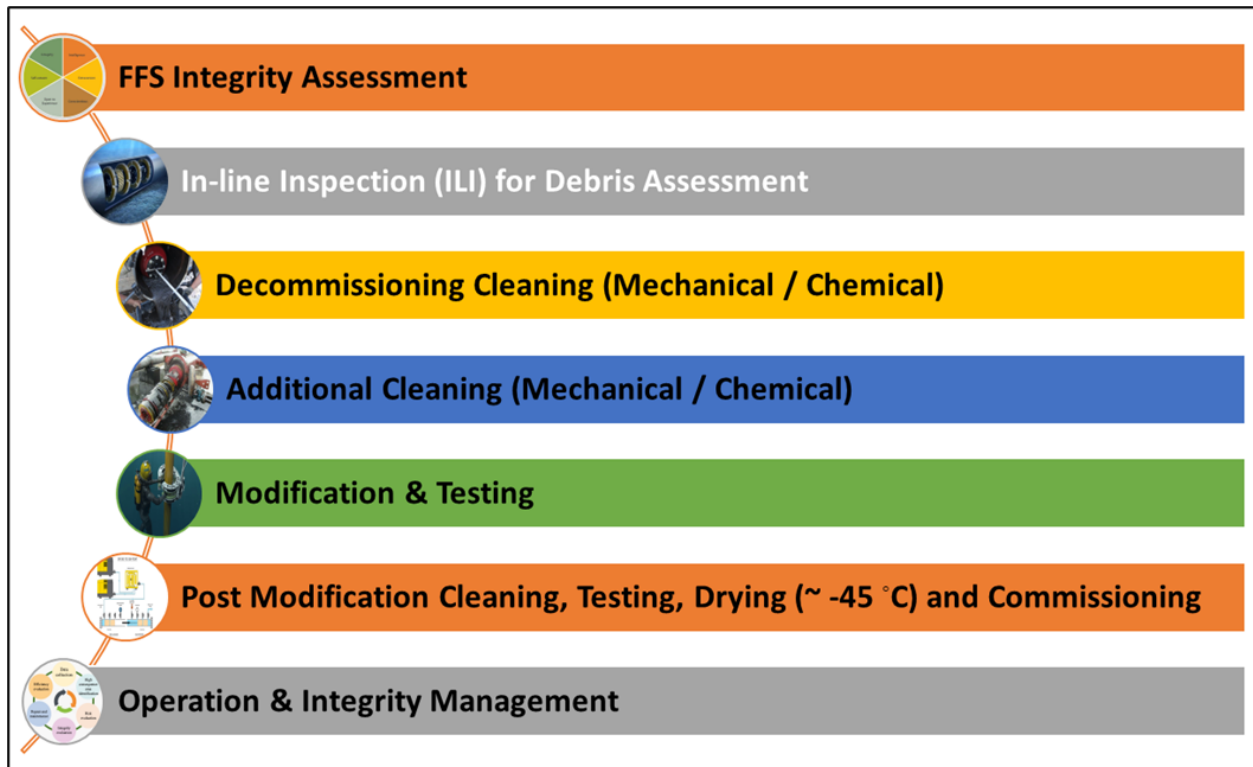
2.1 REPURPOSING AND MODIFICATION OF THE EXISTING PIPELINES

Repurposing the existing pipelines requires a careful consideration of safety, capacity, and compatibility with current operations. Collaboration between industries, government, and regulators is essential to ensure proper infrastructure usage, emission reduction goals, and regulatory compliance.

The fundamental challenge in repurposing pipelines for CCS lies in adapting these infrastructures to accommodate the unique demands of transporting captured carbon dioxide (CO₂) safely and efficiently. As existing pipelines were primarily designed for other purposes, such as natural gas or oil transportation, ensuring their structural integrity becomes paramount when repurposing them for the intricate task of handling and transporting CO₂.

The process shown in Figure 4 will be followed during the repurposing and modification of the existing LBA pipelines for CO₂ service.

Figure 4 – Re-purposing and modification of the existing LBA pipelines



The following existing Eni UK Liverpool Bay Pipelines have been selected for the repurposing and modification of the existing LBA pipelines for CO₂ service:

- PL 1030, 20-inch Gas Export from Douglas to Point of Ayr
- PL 1035, 16-inch Gas Export from Lennox to Douglas
- PL 1036A, 12-inch Gas Production from Lennox to Douglas
- PL 1039, 20-inch Gas Export from Hamilton to Douglas
- PL 1041, 14-inch Gas Export from Hamilton North to Douglas
- P 852, 24-inch Sales Gas from Point of Ayr to Connah's Quay

2.2 FFS INTEGRITY ASSESSMENT

The Inline Inspection data of the existing pipeline system is used to assess the current pipeline Fitness for Service (FFS) integrity and provide the foundation for the re-purposing for CO₂ service.

MFL technology use allows comparison more directly with the historical data from previous inspections. However, UT (ultrasonic) requires a liquid couplant with risks of water contamination. This is where MFL technology steps in, offering a robust and precise solution for the comprehensive inspection of pipeline integrity. MFL technology utilizes magnetic fields and sensors to detect and analyse anomalies within the pipeline's structure that could compromise its integrity. This high level of sensitivity is particularly crucial in the context of repurposing pipelines for CCS, where any compromise in structural integrity could have significant environmental and safety implications. By employing MFL inspections, operators gain a detailed understanding of the pipeline's condition,

enabling them to identify potential weaknesses and proactively address them before they escalate into critical issues.

A pertinent reference supporting the efficacy of MFL technology in pipeline integrity assessment is found in the technical paper titled “Advanced Magnetic Flux Leakage Technology for Pipeline Inspection” by Smith et al. (2018). The paper underscores the precision and reliability of MFL technology in detecting and characterizing anomalies within pipelines, emphasizing its role in ensuring the safe and efficient operation of critical infrastructure. Smith et al.’s work provides valuable insights into the application of MFL technology in diverse pipeline environments, reinforcing its relevance and effectiveness in the evolving landscape of CCS pipeline transportation.

Furthermore, the non-invasive nature of MFL inspections aligns seamlessly with the goal of repurposing existing pipelines for CCS. Without requiring pipeline shutdowns or physical interventions, MFL technology facilitates thorough assessments while minimizing operational disruptions. This non-disruptive approach is essential in the context of CCS, where the efficient and continuous operation of transportation pipelines is integral to the success of carbon capture initiatives.

2.3 IN-LINE INSPECTION (ILI) FOR DEBRIS ASSESSMENT

Through assessment and evaluation during ‘Life of Field’ operations and as part of normal maintenance pigging operations, the cleanliness of each pipeline will be established. This will determine whether additional mechanical cleaning and/or advanced chemical cleaning is required to meet the stringent levels of cleanliness specified by DNVGL-RP-F104 [3].

The Inline Inspection (ILI) programme in 2020/2023 has identified residual solid debris in the six gas pipelines that will be repurposed for CO₂ service. Residual debris will allow local water hold-up in the pipeline. It is critical that all residual debris and water is removed. Pipeline cleaning and drying techniques must achieve a very low dew point. The e-Splora (proprietary Eni technology), video inspection tool and other available techniques will be utilised to evaluate the quality of the pipeline cleanliness for CO₂ service. The e-Splora is a suite of advanced technologies developed by ENI for pipeline inspection and maintenance. It includes various smart devices designed to detect and localize anomalies in pipelines, ensuring their integrity and reducing maintenance costs. The e-Splora multiparametric sensor can be deployed directly into pipelines by attaching it to pigs to collect data on pressure, temperature, and other critical parameters.

2.4 CLEANING (MECHANICAL/CHEMICAL)

Introducing CO₂ into a pipeline network poses several threats to carbon steel pipelines, particularly when any of the following contaminants are also present:

- Water Content
- Existing Debris
- Corrosion
- Subsequent Corrosion Products
- Pitting Corrosion

2.4.1. MECHANICAL CLEANING

The Inline Inspection tool (UT or MFL) will not accurately identify surface deposits, but specialist tools such as e-Splora can be installed onto mechanical cleaning pigs which can then detect hard contact

layer deposits to ensure all such deposits are removed. In addition, camera inspection will also be used to confirm that all contact deposit layers have been removed.

Initial mechanical cleaning will be carried out during decommissioning flushing and cleaning (F&C) operations. These mechanical cleaning operations will be performed using BiDi brush pigs propelled through the pipeline by either filtered treated seawater or nitrogen. It is anticipated that multiple brush pig train runs will reduce the hydrocarbon content to below the decommissioning target Oil in Water concentration (30 ppm). Sampling and analysis of the discharge water and debris will be performed to confirm the required Oil in Water acceptance level and any Additional Mechanical Cleaning requirements. (Additional Mechanical Cleaning will be considered if abnormal types or significant amounts of debris are removed during decommissioning operations). Assessment and evaluation for a satisfactory performance of the cleaning operation will include:

- Contact Deposit Layer Assessment deemed from ILI inspection and Caliper pigging reports and includes Hydrocarbon and Scale deposits.
- Debris Assessment through operational pigging and the sampling/analysis of the debris and foulants recovered during operational pigging.
- Visual or camera inspections to determine the effectiveness of the cleaning operation.

In order to aid the selection of specialist mechanical cleaning tools, debris will be quantified, and samples analysed in a laboratory to determine composition. The laboratory data will help determine which mechanical cleaning tools would be most effective and the most appropriate.

Additional Mechanical Cleaning to achieve a 'Brush Off' level will be achieved by launching tailored pig trains to ensure mechanical cleaning of the pipe wall surface. Continuous assessment of recovered debris and the condition of the pigs themselves will allow for progressive assessments to be made.

This will be followed by successive pig runs until (i) 'no debris' is present in the pig receiver barrel after visual inspection, (ii) the magnets installed on the pigs are free of ferrous debris and (iii) no foulants are found on the pig bodies.

Satisfactory performance of the mechanical cleaning operations will be based on the progressive assessment and evaluation of the amount of received debris and the pig conditions, otherwise chemical cleaning will be considered.

2.4.2. CHEMICAL CLEANING

In order to aid the selection of advanced cleaning chemicals, debris samples must be analysed in a laboratory to determine composition. The laboratory data will help determine which cleaning agent will be the most effective and the most appropriate.

Certain scales are difficult to dissolve (e.g., barium sulphate) and if heavy hydrocarbons such as asphaltenes are adhered to the pipe wall, these may require strong solvents and extended soak times.

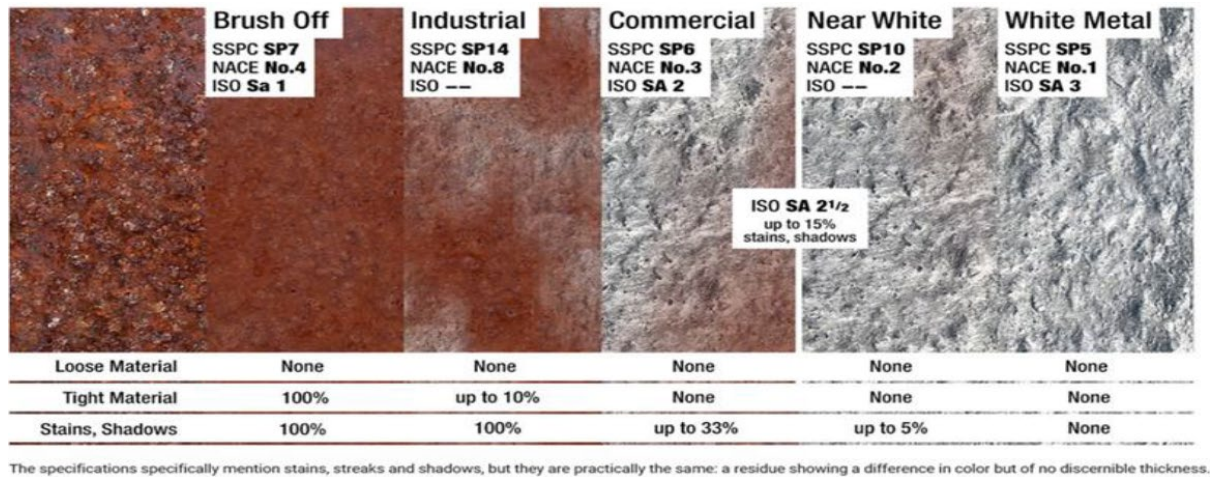
The selection of specialist chemical quantities with specified concentrations must be able to remove debris and foulants adhered to the pipe wall or contact deposit layers e.g. Acid solutions, Alkaline solutions, Chelating agents, Neutralising/Passivating solution, Inhibitors.

The pipeline system cleanliness can be accepted if:

- The pipeline internal surfaces have a uniform clean metallic colour.
- Accessible internal surfaces, when viewed without magnification, shall be free of visible metal deposit, rust, corrosion, oxides or other foreign material.
- A visual or camera inspection confirms the effectiveness of cleaning operation.

The levels of cleanliness required for CCS re-purposed pipelines should be somewhere between Industrial (SSPC SP14 / NACE No. 8) and Commercial (SSPC SP6 / NACE No. 3) as shown in Figure 5.

Figure 5 – Acceptance Criteria for cleanliness required for CCS re-purposed pipelines



On a successful cleaning of the re-purposed pipelines, the following activities will be undertaken:

- Modification and testing,
- Post modification cleaning and local testing,
- Drying (~ minus 45degC),
- Complete pipeline system Integrity Testing
- Commissioning,
- Baseline In-Line Inspection.
- Operation and Pipeline Integrity Management.

3 OPERATION & PIPELINE INTEGRITY MANAGEMENT (PIM)

In the presence of water, CO₂ can form corrosive carbonic acid, presenting potential integrity risks to steel pipelines. It is critical to ensure CO₂ is dehydrated prior to entry into the steel pipelines and monitor the corrosion in the pipelines and manage their integrity.

CO₂ is denser than air. If accidentally released, it can accumulate in open low-lying areas and confined spaces, presenting a toxic hazard and an asphyxiation risk. Air dispersion analysis must be performed to identify CO₂ migration areas in the highly unlikely event of a CO₂ release.

It is very important that Operators of CO₂ pipelines fully understand the hazards, mechanisms, consequences, and probabilities of pipeline failures in order to ensure their safety. There are uncertainties around the conveyance of dense or supercritical phase CO₂ in pipelines, therefore, the current Eni UK LBA approach is to treat CO₂ pipelines as a major accident hazard pipeline.

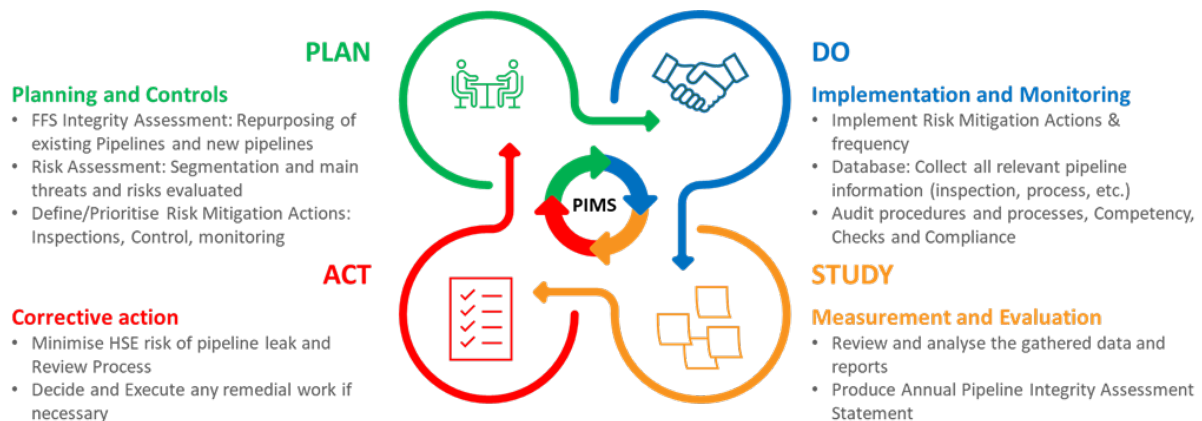
Pipeline operators use various strategies, including monitoring and maintenance programs to detect and address corrosion and other damage during the design life. Magnetic flux leakage (MFL)

inspection can be utilised to mitigate these threats. This paper considers MFL technology use, supporting the repurposing and future integrity, monitoring, and maintenance of the new and existing pipelines.

The current integrity management practice for hydrocarbon pipelines can be applied to the CO₂ pipelines. Pipeline integrity management (PIM) requires planning, implementation, measurement and evaluation, and continuous improvement over the entire lifecycle of the pipeline, with a clear understanding of the design and operating parameters, risks and reliability, inspection requirements, and technical assurance assessment for continued operation. The below steps of the PIM process defined in Figure 6 are as follows:

- Planning and Controls
- Implementation and Monitoring
- Measurement, Evaluation
- Corrective action (Maintenance/Repair)

Figure 6 - Acceptance Criteria for cleanliness required for CCS re-purposed pipelines



The monitoring of PIMS is a live process to assess, monitor and mitigate risk

The Integrity Risk Assessment (Foundation and Operation) process is used to identify all credible threats to pipeline integrity during Project Execution (Decommissioning, Preservation, Modification, Testing, Commissioning and Operation), and assess risk for all threats applicable to the pipeline based on the barriers, likelihood, and potential consequence. The Integrity Risk Assessment process relies on the availability of a wide range of design and operational pipeline information and the experience of the risk assessment team.

The Integrity Risk Assessment will be carried out during the design phase of the project and will be updated immediately after transition to operations. The output from the IRA process is used to define and prioritise actions and integrity management programmes. These include Continuous improvement to PIM and identify requirement for more detailed data to close gaps in information and to reduce uncertainty.

Monitoring is the measurement and collection of data that indirectly gives information on the condition of a component or a system. The monitoring data is typically either through on-line measurements or offline measurements (planned).

Emergency Response Plan: initiating a new rulemaking to update standards for CO2 pipelines, including requirements related to emergency preparedness, and response.

4 PIPELINE INCIDENTS (INCLUDING CO2 & THEIR CAUSES)

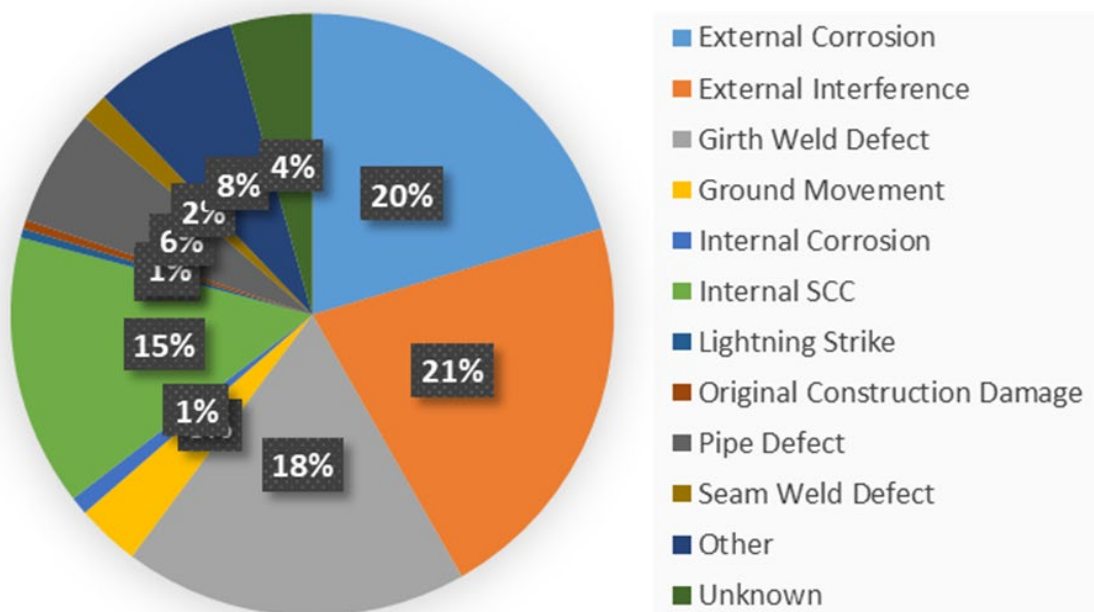
CO2 pipeline incidents have occurred worldwide, comprehensive global data is not readily available. UK and USA pipeline incidents, including CO2, are discussed in the sections below, highlighting the importance of stringent safety protocols and continuous monitoring to ensure the safe transportation of CO2.

4.1 PIPELINE (OIL, GAS OR CHEMICAL) INCIDENTS IN THE UK

Onshore Pipelines

The 2022 Infringement Report [8] shows that landowners were the most likely cause of damage to pipelines in the UK. Of the 316 incidents recorded in 2022 Infringement Report, 40% were caused by landowners. Contractors were also a cause for concern, making up 35% of infringements. An infringement can be someone simply working near an oil, gas or chemical pipeline, without the owner's awareness and permission, through to a worker damaging a pipeline. Figure 7 provides a more detailed breakdown of the various Onshore Pipeline Incidents and Causes.

Figure 7 - Onshore Pipeline Incidents and Causes [8]



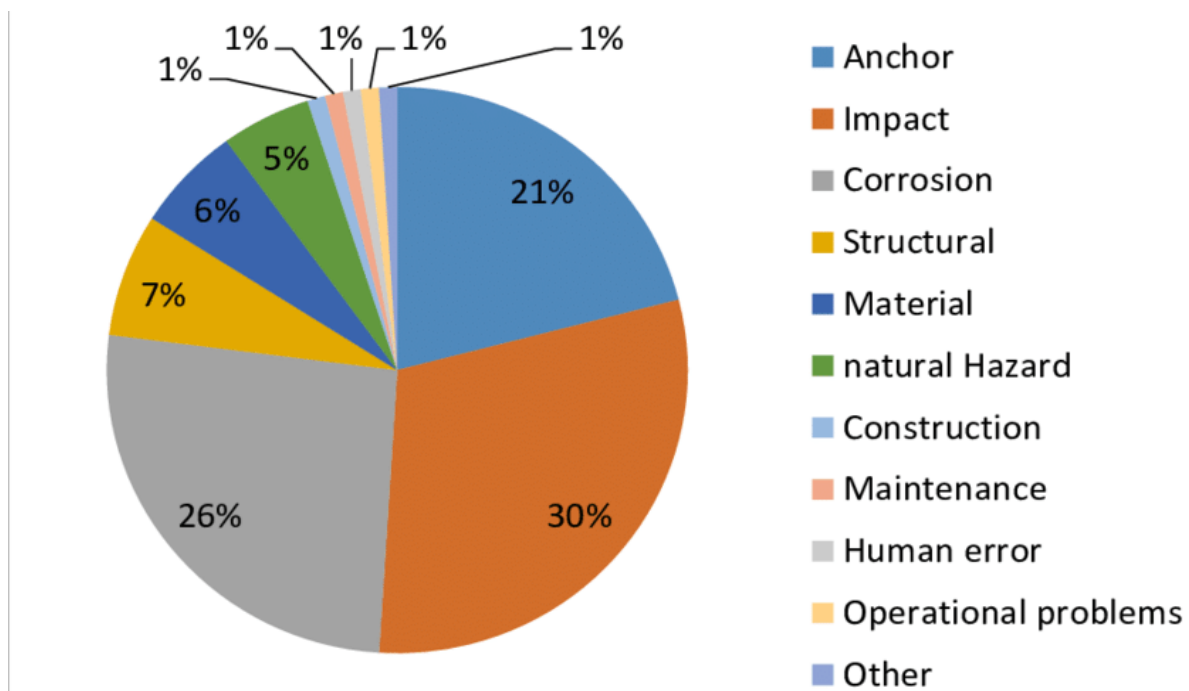
Offshore Pipelines

The Offshore Pipeline and Riser Loss of Containment report (PARLOC) provides details of the offshore pipeline incidents that include all leak incidents from pipelines and risers that have occurred in the United Kingdom Continental Shelf (UKCS) but excludes leak incidents that have occurred in the oil and gas industries of other countries. PARLOC has been updated many times in 2001, 2012 and 2020.

The statistical data from PARLOC 2001 (Figure 8), indicates a significant threat to underwater pipelines during the years 1960 - 2001 in the North Sea. PARLOC 2001 groups the most common causes of damage to the offshore pipelines (Figure 8):

- Damage by anchors of vessels passing above the pipeline
- Impact from a ship hull
- Corrosion
- Structural/technical defects
- Defects in materials
- Natural hazards
- Maintenance
- Human error
- Operational problems, others.

Figure 8 - Offshore Pipeline Incidents and Causes [5]



4.2 PIPELINE (OIL, GAS OR CHEMICAL) INCIDENTS IN THE USA

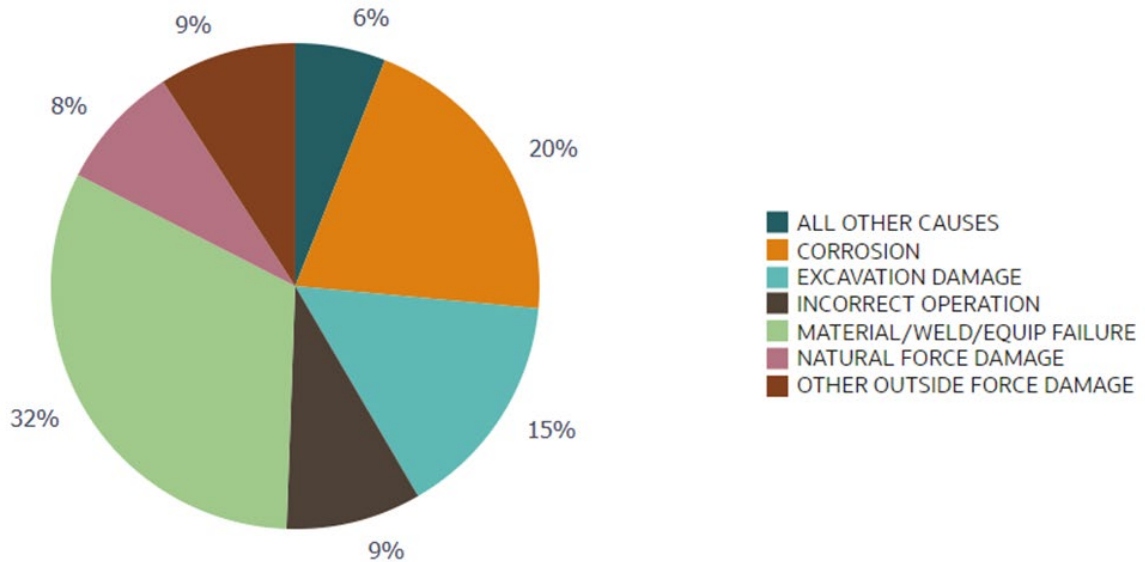
The PHMSA (Pipeline and Hazardous Materials Safety Administration) database [9] provides detailed statistics and trends on pipeline incidents in USA, including those involving CO₂.

PHMSA defines Significant Incidents as any of the following conditions:

- Fatality or injury requiring in-patient hospitalization,
- \$50,000 or more in total costs, measured in 1984 dollars,
- Highly volatile liquid releases of 5 barrels or more, or other liquid releases of 50 barrels or more,
- Liquid releases resulting in an unintentional fire or explosion.

PHMSA data shows [Figure 10] pipeline (oil, gas, or chemical) Significant Incidents trends over the past 20 years (2004 - 2023) which includes serious incidents (those involving fatalities or hospitalizations) and those causing substantial damage or large releases.

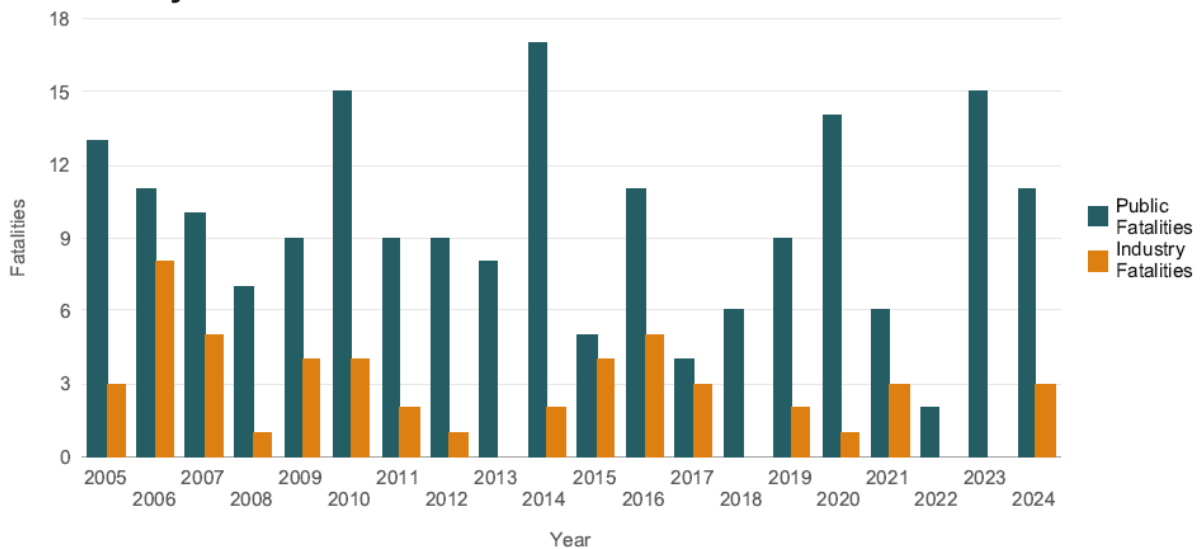
Figure 9 - PHMSA Significant Pipeline Incident Cause Breakdown, 20 Years (2004 - 2023)



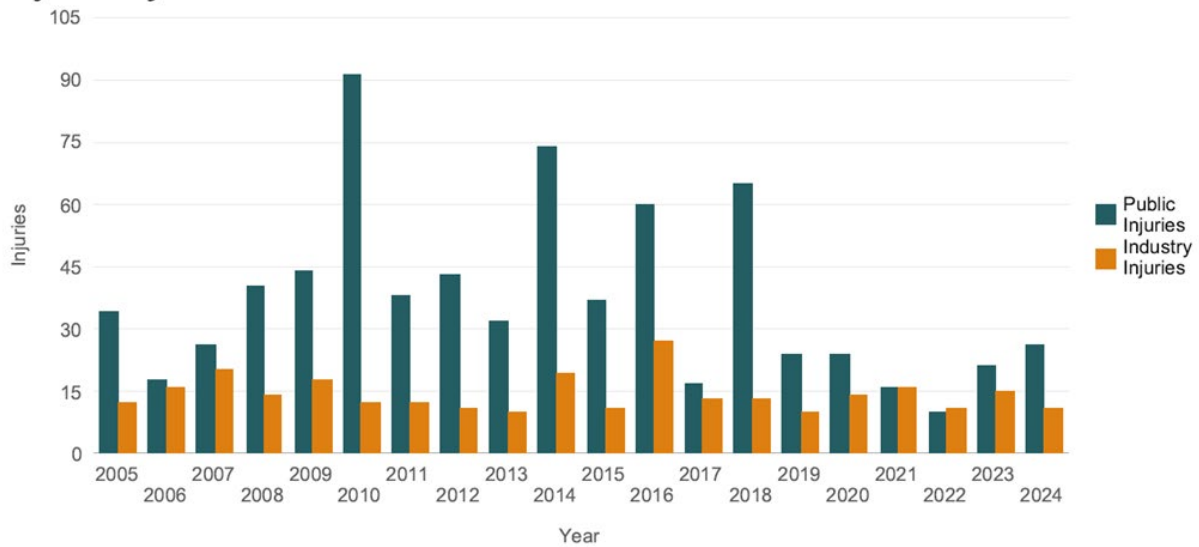
A total of 5773 Significant Incidents were reported in USA related to pipelines (oil, gas, or chemical) between 2004 and 2023 [9]. The significant pipeline incident causes are categorised as corrosion (20%), excavation damage (15%), incorrect operation (9%), material/weld/equipment failure (32%), natural force damage (8%) and other outside force damage (9%). These help in understanding the primary factors contributing to pipeline incidents and the consequences of incidents (Figure 10), detailing the number of fatalities, injuries, and the financial impact on both the public and the pipeline industry.

Figure 10 - PHMSA Pipeline (oil, gas, or chemical) Incident Consequences, 20 Years (2005 - 2024)

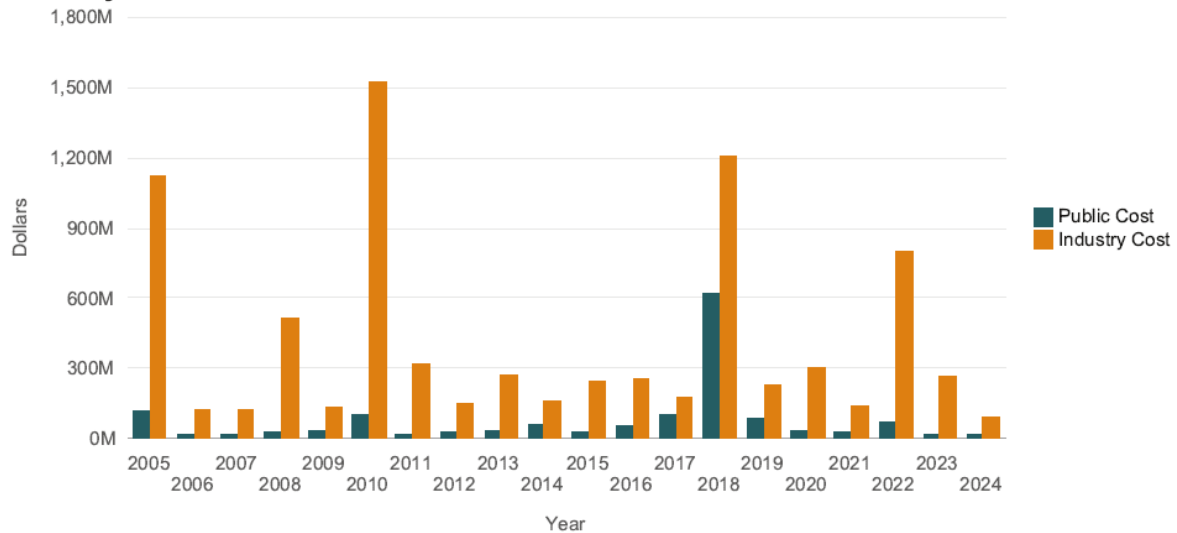
Fatalities by Year



Injuries by Year



Cost by Year



4.3 CO₂ PIPELINE INCIDENTS IN THE USA

The CO₂ pipeline network in the US exceeds 5,000 miles but is far surpassed by the country's nearly 260,000 mile hazardous liquid pipeline network and nearly 3-million-mile natural gas pipeline network [11].

PHMSA data (Figure 11) shows CO₂ pipelines Significant Incident trends over the past 20 years.

A total of 53 Significant Incidents were reported in the USA related to CO₂ pipelines between 2004 and 2023 [9]. The significant incident causes are categorised as corrosion (19%), excavation damage (2%), incorrect operation (6%), material/weld/equipment failure (68%), natural force damage (2%) and other outside force damage (2%). These help in understanding the primary factors contributing to pipeline CO₂ incidents and the consequences of incidents (Figure 12), detailing the number of fatalities, injuries, and the financial impact on both the public and the pipeline industry.

Figure 11 - PHMSA Significant CO2 Pipeline Incident Cause Breakdown, 20 Years (2004 - 2023)

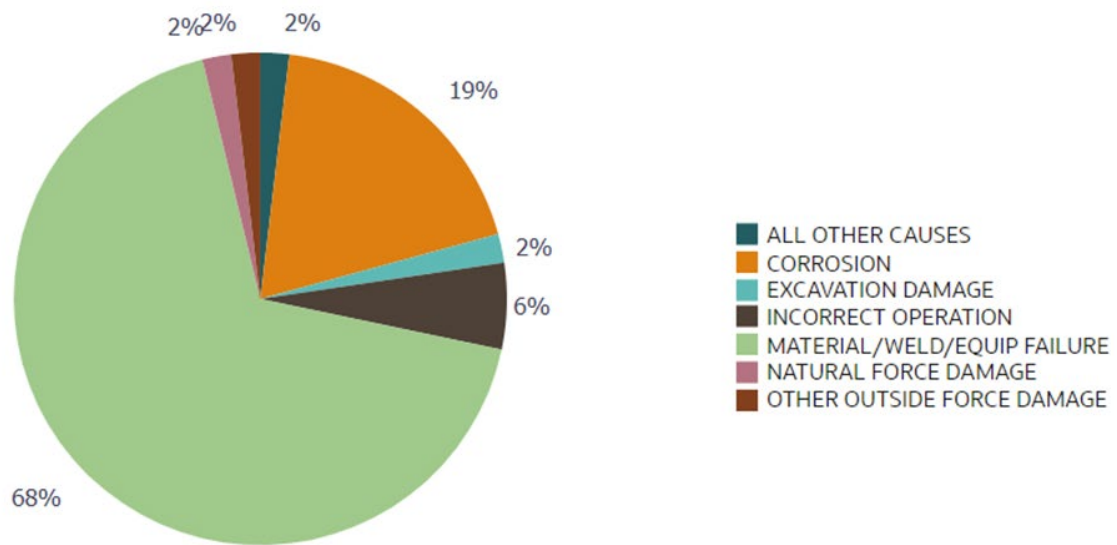
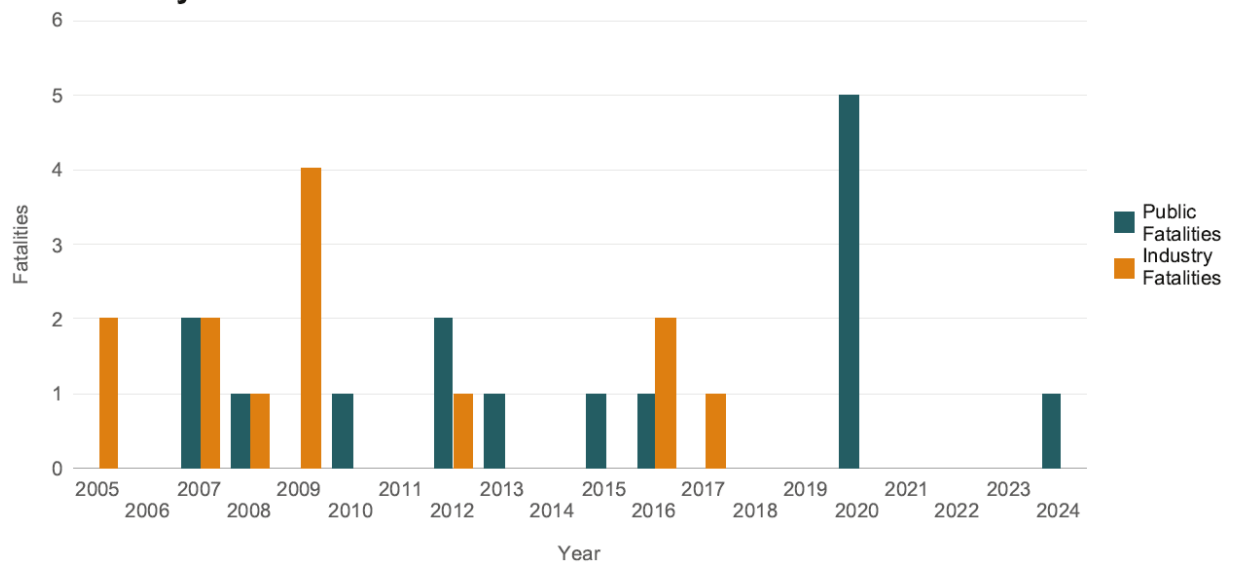
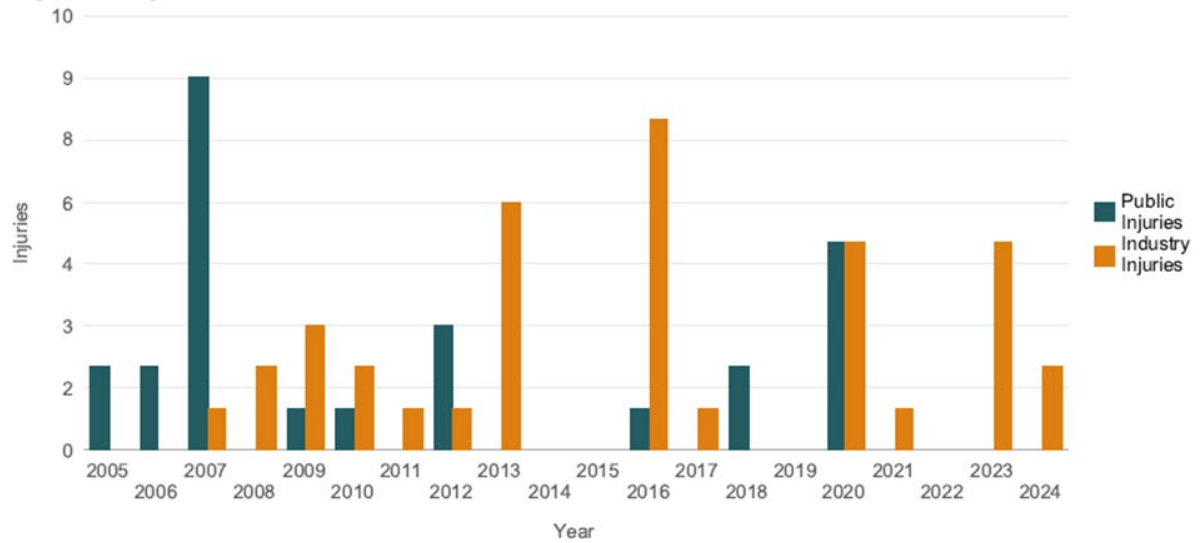


Figure 12 – PHMSA CO2 Pipeline Incident Consequences, 20 Years (2005 - 2024)

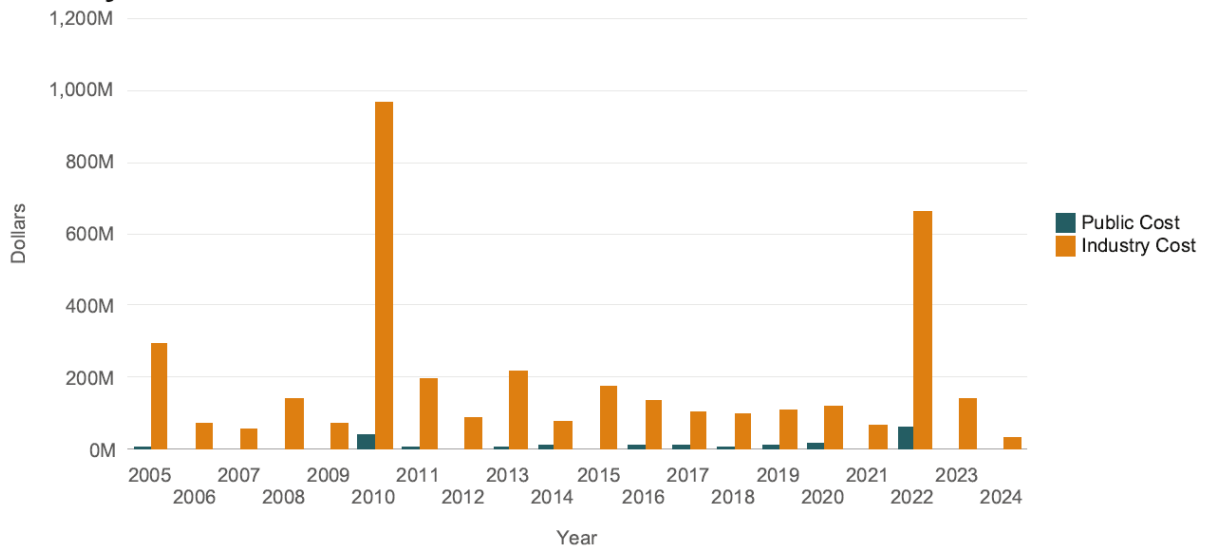
Fatalities by Year



Injuries by Year



Cost by Year



5 INTRODUCTION TO MFL IN CCS (CARBON CAPTURE & STORAGE)

The energy sector has changed dramatically since the coronavirus, while Russia's invasion of Ukraine in February 2022 fuelled an energy crisis and stoked concerns over future energy security. The UK Government responded with support for green jobs and to accelerate our path to net zero. Following this, in March 2023 the UK Government announced £20 billion funding over the next 20 years for CCS projects in the UK as well as the Net Zero Hydrogen Fund (NZHF) and the Hydrogen Business Model (HBM) [10].

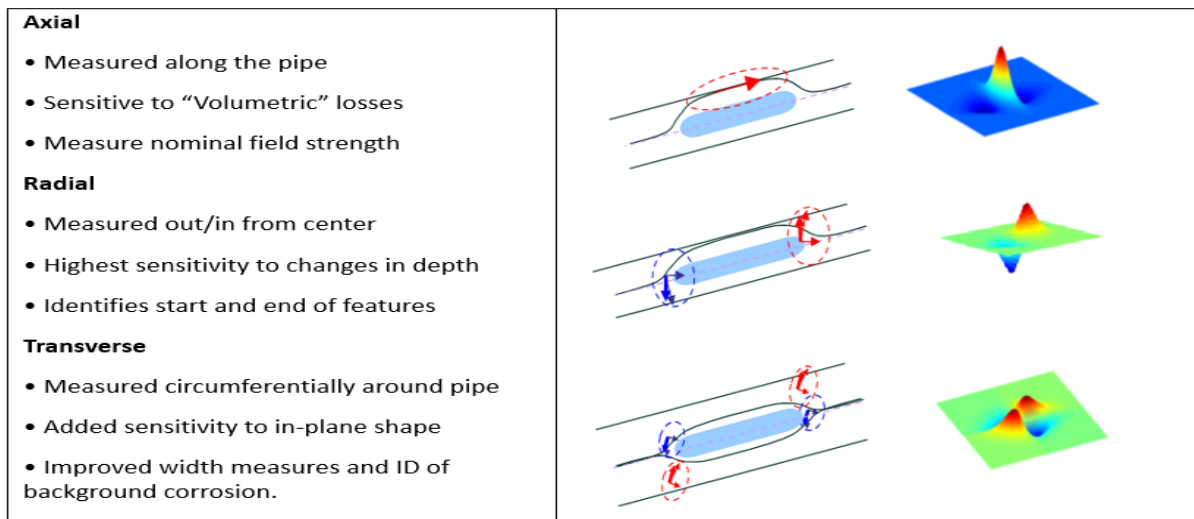
5.1 MFL TECHNOLOGY

Magnetic Flux Leakage (MFL) technology has undergone significant development in the realm of inline pipeline inspection, revolutionizing the way we assess the integrity of pipelines. The Baker Hughes MagneScan™ inspection vehicle has evolved to incorporate advanced detection and sizing capabilities when characterising defects in metallic pipelines.

Initially used for surface inspection, MFL transitioned to inline applications to address the limitations of traditional methods. Early MFL tools consisted of magnetic sensors attached to a pipeline inspection device, which, when passed through the pipe, generated a magnetic field to saturate the material. As the device moved along the pipe, any discontinuities or defects would cause magnetic flux leakage, allowing for the identification of potential issues.

Magnetic flux leakage is affected in multiple vectors and whilst traditional technologies have been developed solely on the axial sensors which measure the flux leakage in a single plain, the MagneScan triaxial sensors detect and record three plains of the leakage: Axial, Radial and Transverse.

Figure 13 - What is recorded and displayed for each component of the magnetic flux leakage vector



These multiple datapoints provide an additional and robust means to identify corrosion profiles and interactions that in many cases are not detectable or distinguishable when using only a single axis inspection tool – regardless of the resolution an alternative inspection vehicle may have. The MagneScan inspection vehicle was selected by ENI to detect and size complex interacting corrosion defects as defined and categorised by POF which could be present within ENI’s UK LBA Pipeline System. Collecting and analysing data recorded by all three vectors allowed Baker Hughes to provide a greater understanding of the morphology of these features.

Miniaturisation and modular design have made MFL tools more versatile and adaptable to various pipeline configurations. Smaller, more agile inspection vehicles can navigate challenging pipeline systems, including tight bends and complex geometries, ensuring comprehensive coverage of the pipeline system.

5.2 RISKS & OPPORTUNITIES USING MFL IN CO2 PIPELINES

1. **Opportunity:** Early Detection of Corrosion

MFL inspection tools excel at identifying corrosion in pipelines. Early detection allows for proactive maintenance, minimizing the risk of CO₂ leakage due to corroded areas.

2. **Risk:** Limited Sensitivity to Internal Pipeline Deposits

MFL may face challenges in detecting internal deposits like solidified CO₂, reducing sensitivity to certain types of anomalies. This limitation poses a risk as it may lead to overlooking potential obstructions or build-up.

3. **Opportunity:** Integration with Advanced Analytics

Leveraging advanced data analytics with MFL can enhance anomaly detection accuracy. Machine learning algorithms can differentiate between benign irregularities and critical defects, improving overall pipeline integrity assessment.

4. **Risk:** High Pressure and Temperature Environments

CO₂ pipelines often operate under high pressure and temperature conditions. MFL tools must be designed to withstand such harsh environments, and failure to do so poses a risk of equipment malfunction and inaccurate inspections.

5. **Opportunity:** Real-time Monitoring for Immediate Response

MFL technology, when coupled with real-time data transmission, offers the opportunity for continuous monitoring. This enables operators to respond promptly to emerging issues, preventing potential leaks and ensuring the safety of CO₂ transport.

MFL inspection tools present opportunities for early corrosion detection, integration with advanced analytics, and real-time monitoring. However, risks include limitations in sensitivity to internal deposits, challenges in high-pressure environments, and the need for robust equipment design to ensure accurate inspections in CO₂ pipelines.

The introduction of carbon dioxide (CO₂) to carbon steel pipelines poses a threat due to the potential for corrosion. When CO₂ dissolves in water, it forms carbonic acid, which can react with the steel, leading to corrosion and degradation of the pipeline over time.

Corrosion in carbon steel pipelines occurs through electrochemical reactions involving the metal surface, water, and corrosive agents like carbonic acid. This process accelerates in the presence of dissolved CO₂, and it can result in the formation of corrosion products such as iron carbonate, which can compromise the structural integrity of the pipeline.

Such a change in pipeline media drives a requirement to understand the condition of the pipeline in terms of corrosion threats or indeed active corrosion present – this baseline prior to change of service allows for a clear comparison and greater understanding of the impact of the product on the pipeline.

The original plan below highlights the stakeholder involvement in the provision of services for the ILI of ENI's offshore pipelines within the CCS project.

5.3 CASE STUDY: ENI ILI PROGRAMME FOR RE-PURPOSING– 2022/23

Eni UK LBA Pipelines System and Pigging History

The installations are connected by pipelines, which are predominantly in burial on the seabed. Each pipeline has been routinely sphered and/or pigged during operations to date.

Each pipeline has a fully functional pig launcher and pig receiver. These were designed for the launch and receipt of intelligent pigs.

The 12" diameter gas pipeline pig receiver at Douglas is oriented as a pig launcher. It required a pig receiver insert to allow receipt of inspection pigs. This insert was provided by Eni UK.

All gas pipelines have been routinely inspected by magnetic flux leakage (MFL) intelligent pig and had been inspected previously by Baker Hughes. The interval between previous inspections has been mostly every five years except for the onshore gas pipeline which is a sales gas pipeline.

As a fullstream support services business, Baker Hughes undertook the following scopes of work for the four pipelines:

- Flooding, Cleaning and Gauging operations
- MFL inspection operation
- Dewatering operation
- Reinstatement testing of the topside pipework

Worksite locations are Douglas Platform, Satellite platforms (Hamilton North/ Lennox/Hamilton) and on the deck of the support jack- up platform, The Irish Sea Pioneer.

6 CONCLUSION & RECOMMENDATIONS

Introducing CO₂ into pipelines can damage carbon steel pipes, especially if there are contaminants like water, debris, and corrosion. To reuse these pipelines safely, we need to carefully consider safety, capacity, and how well they fit with current operations. It's important for industries, government, and regulators to work together to use the infrastructure properly, meet emission reduction goals, and follow regulations.

Accurate and reliable inline inspection data helps assess the current condition of pipelines and determine if they can be repurposed for CO₂ transport. During regular maintenance and operations, the cleanliness of each pipeline is checked. This will show if more mechanical or chemical cleaning is needed to meet strict cleanliness standards. The CCS project will use various technologies like MFL (using tri-axial sensors), e-Splora, and video inspection to evaluate pipeline condition. If pipelines aren't properly cleaned and dried before use, it can cause problems. Therefore, thorough cleaning and drying are essential steps to reach the very low dew point needed for CO₂ transport.

To choose the right mechanical and chemical cleaning methods, debris will be measured and analysed in a lab to identify its composition. This helps in selecting the appropriate cleaning tools and agents. Effective cleaning will be based on ongoing assessments. The levels of cleanliness required for CCS re-purposed pipelines should be somewhere between Industrial (SSPC SP14 / NACE No. 8) and Commercial (SSPC SP6 / NACE No. 3) as shown in Fig 5.

After successfully cleaning the repurposed pipelines, several key activities will be carried out:

- Modification and testing
- Post-modification cleaning and local testing
- Drying (around -45°C)

- Complete pipeline system integrity testing,
- Commissioning
- Baseline in-line inspection
- Operation and pipeline integrity management.

These steps ensure the pipelines are ready and safe for CO₂ transport.

When water is present, CO₂ can form corrosive carbonic acid, which poses integrity risks to steel pipelines. Therefore, it's crucial to dehydrate CO₂ before it enters the pipelines and to monitor and manage corrosion. Operators of CO₂ pipelines need to fully understand the hazards, mechanisms, consequences, and probabilities of pipeline failures to ensure safety. Due to uncertainties in transporting dense or supercritical phase CO₂, the current approach by Eni UK LBA is to treat CO₂ pipelines as major accident hazard pipelines.

MFL inspection tools employing tri-axial sensor capability can detect complex corrosion early, integrate with advanced analytics, and provide real-time monitoring. However, they have some limitations, such as reduced sensitivity to internal deposits and the need for strong equipment design to ensure accurate inspections in CO₂ pipelines.

The integrity management practices used for hydrocarbon pipelines can also be applied to CO₂ pipelines. Pipeline Integrity Management (PIM) involves planning, implementation, measurement and evaluation, and continuous improvement throughout the pipeline's lifecycle. This requires a clear understanding of the design and operating parameters, risks and reliability, inspection requirements, and technical assurance assessments to ensure ongoing safe operation.

Learning from UK and USA pipeline incidents, including CO₂, shows the importance of strict safety protocols and continuous monitoring to ensure safe CO₂ transport and early warning systems. Between 2004 and 2023, 53 significant CO₂ pipeline incidents were reported in the USA. The causes were:

- Corrosion (19%)
- Excavation damage (2%)
- Incorrect operation (6%)
- Material/weld/equipment failure (68%)
- Natural force damage (2%)
- Other outside force damage (2%)

These incidents help us understand the main factors leading to CO₂ pipeline issues and their consequences, including fatalities, injuries, and financial impacts on the public and the pipeline industry.

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